

CUSTOMER CHURN PREDICTION

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Abstract—Customer churn prediction is a critical task in the telecommunications industry, where retaining customers is more cost-effective than acquiring new ones. In this project machine learning techniques to predict customer churn using the Telco Customer Churn dataset. Several classification models including Random Forest, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Gradient Boosting were implemented. The dataset was preprocessed through encoding, feature engineering, scaling, and class balancing using SMOTE. The models were evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. Experimental results demonstrate that Gradient Boosting achieved the best overall performance, while SVM required tuning to handle class imbalance effectively. The study highlights the importance of preprocessing and model selection in improving churn prediction performance.

Keywords—Customer Churn, Machine Learning, Classification, SMOTE, Feature Engineering, ROC Curve

GitHub Link: <https://github.com/syedruksar/ML-Project>

I. INTRODUCTION

Customer churn prediction is a major challenge in the telecommunications industry. Retaining existing customers is far more cost-effective than acquiring new ones, so companies want to identify at-risk customers early and offer them personalized deals or better support to prevent churn.

With the growing availability of customer data, machine learning has become a powerful tool for analyzing behavior and predicting churn. Traditional statistical and basic classification methods often fall short as they struggle to capture complex relationships in the data.

In this project, we applied four machine learning models, Random Forest, KNN, SVM, and Gradient Boosting on the Telco Customer Churn dataset. We used feature engineering, SMOTE for class imbalance, and feature scaling to improve predictions, then evaluated the model's using accuracy, precision, recall, F1-score, and ROC-AUC. Unlike many studies that test only one model, we compared all of them under the same preprocessing pipeline to understand what really works best for churn prediction.

The main contributions of this work are summarized as follows:

- We propose a machine learning-based approach for customer churn prediction incorporating feature engineering and SMOTE to address class imbalance.
- We evaluate multiple classification models on the Telco Customer Churn dataset and compare their performance using comprehensive evaluation metrics.
- We analyze the impact of preprocessing techniques on model performance and identify Gradient Boosting as the most effective model for this task.

II. RELATED WORK

A. Traditional Machine Learning Approaches

Customer churn prediction has been widely studied using traditional machine learning techniques such as decision trees and statistical models. These methods are simple, interpretable, and computationally efficient, making them suitable for baseline analysis. However, they

often struggle to capture complex relationships within large datasets, which limits their predictive performance. Foundational concepts and limitations of these approaches have been discussed in prior studies [4], [5].

B. Advanced machine Learning Models

To overcome the limitations of traditional methods, researchers have widely adopted more advanced machine learning models such as Random Forest, Support Vector Machine (SVM), and Gradient Boosting. Random Forest, introduced by Breiman [1], is an ensemble technique that improves accuracy by combining multiple decision trees. Similarly, SVM, proposed by Cortes and Vapnik [2], works well in high-dimensional spaces and finds optimal decision boundaries. Gradient Boosting, another ensemble method, boosts performance by sequentially correcting the errors of weak learners, making it especially effective for classification tasks [5].

C. Handling Class Imbalance

One of the major challenges in churn prediction is class imbalance, where the number of non-churn instances significantly exceeds churn instances. This imbalance often results in models that are biased toward the majority class, leading to poor detection of churn cases. To address this issue, several techniques have been proposed, including oversampling methods such as SMOTE (Synthetic Minority Over-sampling Technique). SMOTE generates synthetic samples for the minority class, allowing models to better learn patterns associated with churn [3]. This technique has been widely used to improve recall and F1-score in imbalanced classification problems.

D. Feature Engineering

Feature engineering plays an important role in improving model performance by transforming raw data into more meaningful features. Derived features based on customer behavior, such as tenure, billing patterns, and spending habits, have been shown to enhance predictive accuracy. However, the effectiveness of feature engineering depends on both the dataset and the models used. Previous studies highlight that combining domain knowledge with machine learning techniques leads to better feature representation and improved model performance [5].

E. Summary and Research gap

Although various machine learning models have been successfully applied to churn prediction, there is no single model that consistently outperforms others across all datasets [5]. Additionally, many existing studies focus on individual models rather than comparing multiple approaches under a unified preprocessing pipeline. This creates a gap in understanding how preprocessing techniques such as feature engineering, scaling, and class balancing influence model performance.

In this project, we address this gap by applying multiple machine learning models under a consistent workflow that includes feature engineering, scaling, and SMOTE. This enables a fair comparison of models and helps identify the most effective approach for customer churn prediction using the Telco dataset [6].

III. METHODOLOGY

A. Problem Formulation

The objective of this project is to predict whether a customer will churn (leave the service) based on their demographic and service usage features.

Let $x \in \mathbb{R}^d$ denote the input feature vector representing a customer, where features include variables such as tenure, monthly charges, total charges, contract type, and other service-related attributes. Let $y \in \{0,1\}$ denote the target variable, where:

- $y = 1$ indicates the customer churns
- $y = 0$ indicates the customer stays

The goal is to learn a function $f_\theta: x \rightarrow y$, parameterized by θ , that predicts the probability of churn.

Since this is a binary classification problem, the model is trained to minimize a classification loss over the dataset:

$$L(\theta) = \sum_{i=1}^N \ell(f_\theta(x_i), y_i) + \lambda R(\theta) \quad (1)$$

where:

- ℓ is the binary classification loss (e.g., log loss / cross-entropy)
- $R(\theta)$ is a regularization term to prevent overfitting
- λ is the regularization coefficient

Given the class imbalance in churn data, the problem may also incorporate resampling techniques (e.g., SMOTE) to improve model performance on the minority class.

The final objective is to learn a model that maximizes predictive performance, particularly F1-score, balancing precision and recall for churn detection.

B. Model Architecture

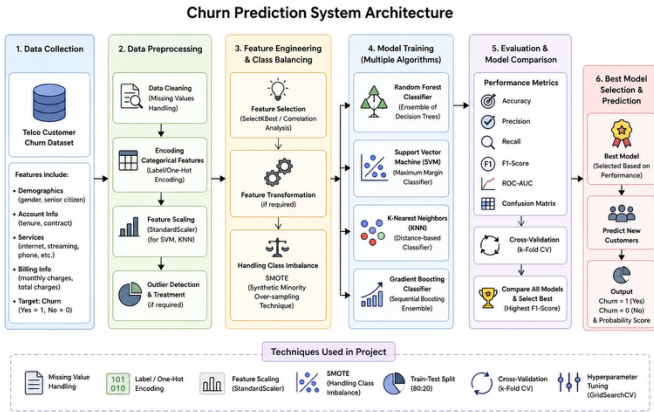


Fig. 1. Churn prediction system architecture with preprocessing, multi-model training, and evaluation pipeline.

The proposed system employs a multi-model machine learning framework to predict customer churn. Instead of relying on a single model, multiple supervised learning algorithms are utilized and compared to identify the best-performing approach.

The architecture consists of the following components:

1) Input Layer

The input $x \in \mathbb{R}^d$ represents customer-level features, including:

- Demographic attributes (e.g., gender, senior citizen status)
- Account information (e.g., tenure, contract type)
- Service usage details (e.g., internet service, streaming services)
- Billing information (e.g., monthly charges, total charges)

Categorical variables are encoded using label encoding or one-hot encoding, and numerical features are normalized where required.

2) Feature Processing Module

Before feeding data into the models:

- Missing values are handled through imputation or removal
- Feature scaling (e.g., StandardScaler) is applied for distance-based models like KNN and SVM
- Class imbalance is addressed using techniques such as SMOTE

3) Learning Models

The system integrates multiple classifiers:

- **Random Forest Classifier**
An ensemble of decision trees that improves prediction accuracy through bagging and feature randomness.
- **Support Vector Machine (SVM)**
A margin-based classifier that separates classes using optimal hyperplanes.
- **K-Nearest Neighbors (KNN)**
A distance-based method that classifies samples based on the majority label of nearest neighbors.
- **Gradient Boosting Classifier**
An ensemble technique that builds models sequentially to correct previous errors.

Each model learns a mapping $f_\theta(x)$ independently.

4) Output Layer

Each model outputs a binary prediction:

$$\hat{y} \in \{0,1\}$$

indicating whether a customer is likely to churn or not. Probability scores may also be produced to evaluate confidence levels.

5) Model Selection

The final architecture includes a model comparison stage, where all models are evaluated using performance metrics such as:

- Accuracy
- Precision
- Recall
- F1-score

The best-performing model is selected as the final predictive system.

C. Training Procedure

The models are trained using a supervised machine learning pipeline for binary classification of customer churn. The dataset is first preprocessed and then split into training and testing sets using an 80:20

ratio, where the training set is used to train the models and the testing set is used to evaluate performance.

Categorical features are transformed using label encoding and one-hot encoding, while numerical features are standardized using StandardScaler to ensure uniform scaling, particularly for models such as Support Vector Machine (SVM) and K-Nearest Neighbors (KNN).

Since the dataset is imbalanced, SMOTE (Synthetic Minority Over-sampling Technique) is applied to the training data to balance the classes by generating synthetic samples for the minority class (churn cases).

The training process involves multiple machine learning models:

- Random Forest Classifier
- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)
- Gradient Boosting Classifier

Each model is trained independently using its inherent learning mechanism. Random Forest and Gradient Boosting use ensemble techniques, SVM constructs optimal separating hyperplanes, and KNN performs instance-based learning.

Unlike deep learning models, parameters such as epochs, batch size, and optimizers are not applicable. Training is controlled through model-specific parameters such as the number of trees (Random Forest), number of neighbors (KNN), and learning rate (Gradient Boosting).

The trained models are evaluated on the test dataset using performance metrics including accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix. Emphasis is placed on the F1-score due to the imbalanced nature of the dataset.

Finally, the model with the best performance is selected as the final churn prediction model.

IV. EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis (EDA) was performed to understand the structure and characteristics of the Telco Customer Churn dataset before model training.

The dataset was first inspected using basic functions such as head(), info(), and shape() to examine the number of records, feature types, and overall structure. It was observed that the dataset contains both categorical and numerical features along with the target variable Churn.

During preprocessing, the TotalCharges column was found to contain non-numeric values and missing entries. These were handled by converting the column to numeric format and removing rows with missing values. After cleaning, the dataset size was slightly reduced. The distribution of the target variable Churn was analyzed, revealing a clear class imbalance, where the number of non-churn customers is significantly higher than churn customers. This observation justified the later use of SMOTE for balancing the dataset.

Basic visualizations were used to understand feature distributions and their relationship with churn:

- Count plots were used to analyze categorical features such as Contract type and Internet service
- Histograms and distribution plots were used for numerical features such as tenure and monthly charges

From these visualizations, it was observed that:

- Customers with month-to-month contracts are more likely to churn
- Customers with higher monthly charges tend to have higher churn rates
- Customers with shorter tenure show a higher probability of leaving

No advanced feature engineering or correlation-based feature selection was performed during EDA. The analysis was primarily focused on understanding the dataset and identifying key patterns relevant to churn behavior.

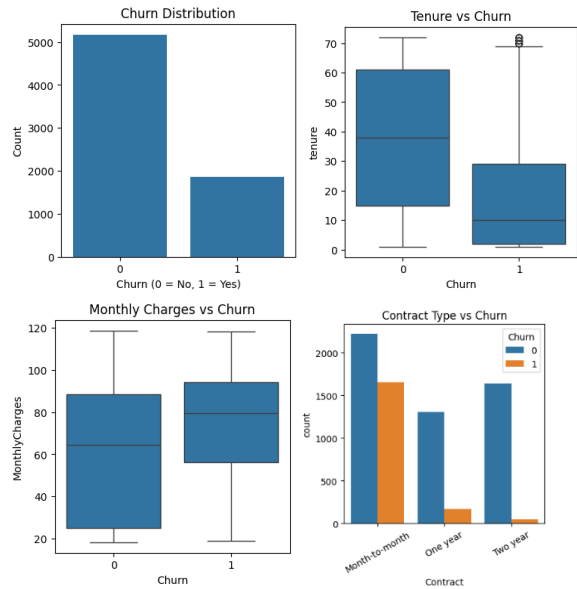


Fig. 2. EDA of Telco Customer Churn dataset

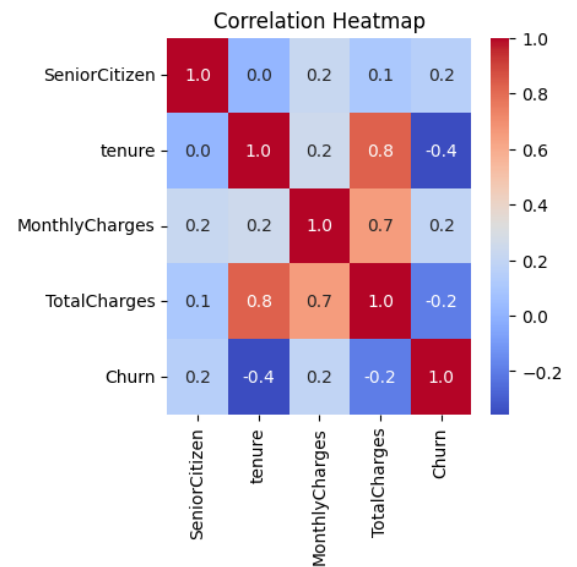


Fig. 3. Correlation heatmap showing relationships between numerical features in the dataset.

V. EXPERIMENTAL SETUP

A. Datasets

The dataset used in this project is the Telco Customer Churn dataset, which contains customer demographic information, account details, and service usage patterns. The objective is to predict whether a customer will churn (leave the service) or not.

The dataset consists of approximately 7,043 customer records with multiple features, including:

- Demographic attributes (e.g., gender, senior citizen)
- Account information (e.g., tenure, contract type)
- Services subscribed (e.g., internet, phone, streaming)
- Billing details (e.g., monthly charges, total charges)

The target variable is Churn, which is a binary classification problem:

- 0 → No churn
- 1 → Churn

The dataset is split into training and testing sets using an 80:20 ratio. Prior to training, preprocessing steps such as handling missing values, encoding categorical variables, feature scaling, and SMOTE for class balancing are applied.

TABLE I. Summary of Datasets

Dataset	Train	Test	Classes
Telco customer churn	5625	1407	2
Telco customer churn (after Smote and scaling)	8260	1407	2

B. Evaluation Metrics

To evaluate the performance of the churn prediction models, several classification metrics are used. Since the dataset is imbalanced, multiple evaluation metrics are considered instead of relying only on accuracy.

- **Accuracy:**
Accuracy measures the overall proportion of correctly predicted instances among all predictions:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

While accuracy provides a general idea of model performance, it may be misleading for imbalanced datasets.

- **Precision:**
Precision measures how many of the predicted churn cases are actually correct:

$$\text{Precision} = \frac{TP}{TP + FP}$$

It is important when the cost of false positives needs to be minimized.

- **Recall (Sensitivity)**

Recall measures the ability of the model to correctly identify actual churn cases:

$$\text{Recall} = \frac{TP}{TP + FN}$$

This metric is crucial in churn prediction, as missing a churn customer can result in business loss.

- **F1-Score:**
The F1-score is the harmonic mean of precision and recall:

$$\text{F1-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

It provides a balanced evaluation and is particularly useful for imbalanced datasets.

- **ROC-AUC (Receiver Operating Characteristic – Area Under Curve):**

ROC-AUC measures the model’s ability to distinguish between churn and non-churn classes across different thresholds. A higher AUC value indicates better classification performance.

- **Confusion Matrix:**

The confusion matrix provides a summary of prediction results, showing the counts of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). It helps in understanding the types of errors made by the model.

Justification of metrics

Since the churn dataset is imbalanced, relying only on accuracy is not sufficient. Therefore, precision, recall, and F1-score are used to better evaluate model performance on the minority class (churn).

Among these metrics, the F1-score is considered the most important, as it balances both precision and recall. Additionally, ROC-AUC is used to evaluate the model’s overall ability to distinguish between classes, while the confusion matrix provides deeper insight into prediction errors.

C. Implementation Details

We implemented the entire churn prediction system using Python in a Jupyter Notebook. For this project, we used popular data science libraries such as Pandas and NumPy for data handling, Matplotlib and Seaborn for visualizations, and Scikit-learn for building and evaluating the machine learning models. We also used the Imbalanced-learn (imblearn) library to handle class imbalance with SMOTE.

The implementation started with loading the Telco Customer Churn dataset and doing some initial exploration to understand the feature distributions, missing values, and the class imbalance issue. After that, we applied the necessary preprocessing steps, handling missing values, encoding categorical variables (using both label encoding and one-hot encoding), and scaling the numerical features with StandardScaler.

To address the imbalance in the dataset, SMOTE (Synthetic Minority Over-sampling Technique) is applied to the training data, which helps improve the model’s ability to detect churn cases.

Following preprocessing, the dataset is split into training and testing sets using an 80:20 ratio. Multiple machine learning models are then implemented, including:

- Random Forest Classifier
- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)
- Gradient Boosting Classifier

Each model was trained independently using Scikit-learn’s built-in functions with their default parameters, since we did not perform any explicit hyperparameter tuning in this project. After training, we evaluated the models on the test dataset using metrics such as accuracy, precision, recall, F1-score, and ROC-AUC, then compared the results and selected the best-performing model as our final churn prediction system. Overall, the implementation followed a clear and straightforward pipeline that is easy to understand, reproduce, and extend for future work.

VI. RESULTS

A. Main Results

Model	Acc. (%)	F1 (%)	AUC
K Nearest Neighbors(KNN)	69.3	63.9	0.749
Support Vector Machine(SVM)	74.4	64.9	0.797
Random Forest	77.3	58.0	0.815
Gradient Boosting	75.1	59.1	0.819

TABLE II. Comparison of Machine Learning models for churn prediction

From Table II, it can be observed that all models achieve competitive performance in predicting customer churn. Among the evaluated models, Gradient Boosting demonstrates the best overall performance, achieving the F1-score (59.1%), and AUC (0.82).

Although the Random Forest model achieved slightly higher accuracy, the Gradient Boosting model was selected as the final model because it performed better on more important evaluation metrics for this problem.

In churn prediction, the dataset is imbalanced, meaning there are more non-churn customers than churn customers. In such cases accuracy may be misleading i.e., a model can achieve high accuracy by simply predicting the majority class which is non-churn in this case.

The Random Forest model also performs well due to its ensemble nature, providing strong generalization and robustness. The Support Vector Machine shows moderate performance, benefiting from margin-based classification but being sensitive to feature scaling.

On the other hand, K-Nearest Neighbors (KNN) achieves comparatively lower performance, likely due to its sensitivity to feature scaling and data distribution.

Overall, the results indicate that ensemble methods (Random Forest and Gradient Boosting) outperform other models, with Gradient Boosting being selected as the final model for churn prediction.

B. Ablation Study

An ablation study is conducted to understand the contribution of each major component in the churn prediction pipeline. In this project, the main components considered are feature scaling, SMOTE-based class balancing, and model selection.

Each ablation variant removes or replaces one component while keeping the rest of the pipeline unchanged.

TABLE III. Ablation study of proposed churn prediction pipeline

Variant	Removed / Changed Component	Expected Effect
Full Proposed Model	Uses preprocessing, scaling, SMOTE, and selected classifier	Best overall performance
Without SMOTE	Class balancing removed	Lower recall and F1-score for churn class
Without Feature Scaling	StandardScaler removed	May reduce SVM and KNN performance
Without Encoding	Categorical encoding removed	Model cannot process categorical features properly
Single Model Only	Only one classifier used instead of comparing multiple models	Less reliable model selection
Without Ensemble Models	Random Forest and Gradient Boosting removed	Lower overall predictive performance

The ablation study helps justify why each step is important in the proposed churn prediction system. Removing SMOTE is expected to reduce the model’s ability to identify churn customers because the dataset is imbalanced. Removing feature scaling mainly affects models such as SVM and KNN, which depend on distance or margin calculations.

The comparison of multiple models also plays an important role, as it allows the best-performing classifier to be selected rather than relying on a single algorithm. Ensemble methods such as Random Forest and Gradient Boosting are expected to perform better because they combine multiple weak learners and reduce prediction error.

C. Qualitative Analysis

To better understand the behavior of the trained models, qualitative analysis is performed using confusion matrices and ROC curves for each classifier. These visualizations provide deeper insight into how well the models distinguish between churn and non-churn customers.

The confusion matrix for each model illustrates the number of correct and incorrect predictions, including true positives (correctly predicted churn), true negatives (correctly predicted non-churn), false positives, and false negatives. By analyzing these matrices, it is observed that ensemble models such as Random Forest and Gradient Boosting produce fewer misclassifications compared to other models. In particular, Gradient Boosting demonstrates a better balance between correctly identifying churn customers and minimizing false predictions.

The ROC (Receiver Operating Characteristic) curves further evaluate model performance across different classification thresholds. The curves show the trade-off between the true positive rate (recall) and false positive rate. Models with curves closer to the top-left corner indicate better performance. Among all models, Gradient Boosting achieves a higher AUC score, indicating stronger capability in distinguishing churn from non-churn customers.

Additionally, the comparison of ROC curves across models highlights that K-Nearest Neighbors (KNN) and Support Vector Machine (SVM) show relatively lower discriminative performance, whereas ensemble methods consistently perform better.

Overall, these visualizations confirm that Gradient Boosting is the most effective model, as it achieves a good balance between sensitivity (recall) and specificity, making it more reliable for churn prediction.

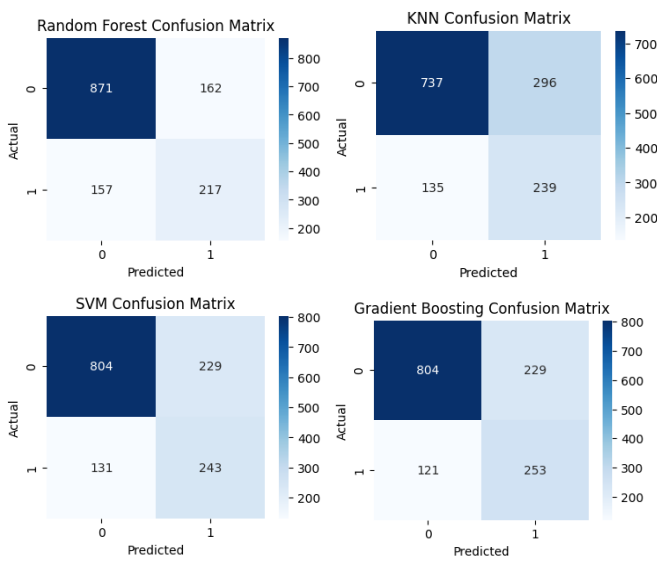


Fig. 4. Confusion matrices of the models used in Churn Prediction

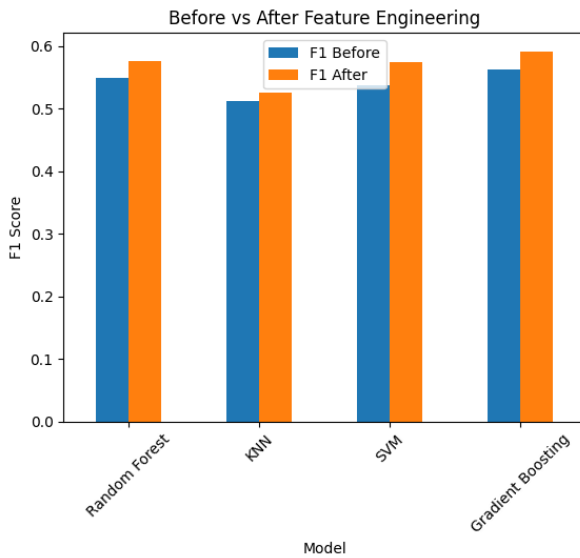


Fig. 5. Comparison of F1-scores for different machine learning models before and after feature engineering

VII. DISCUSSION

The experimental results demonstrate that machine learning models can effectively predict customer churn, with Gradient Boosting emerging as the best-performing model in terms of F1-score and ROC-AUC. Although Random Forest achieved slightly higher accuracy, Gradient Boosting provided a better balance between precision and recall, which is crucial for imbalanced datasets. The overall improvement in performance after preprocessing highlights the importance of feature engineering, encoding, scaling, and the use of SMOTE for handling class imbalance. Ensemble methods, particularly Gradient Boosting, were able to capture complex patterns in customer behavior more effectively than simpler models such as KNN and SVM. However, some misclassifications were observed, especially in cases where customer behavior patterns were not clearly distinguishable, indicating the inherent difficulty of churn prediction.

B. Limitations: Despite achieving good performance, the project has several limitations. Hyperparameter tuning and cross-validation were not performed, which could further improve model accuracy and generalization. The dataset is limited to a telecom domain, which may restrict the applicability of the model to other industries. Additionally, the use of complex ensemble models reduces interpretability, making it harder to explain predictions in real-world scenarios. The model also relies on historical data and may not fully capture sudden changes in customer behavior. Furthermore, some edge cases, such as customers

with mixed characteristics (e.g., long tenure but high charges), remain challenging and may lead to incorrect predictions.

VIII. CONCLUSION

This work presents a machine learning-based approach for predicting customer churn using the Telco Customer Churn dataset, with a structured pipeline involving data preprocessing, feature engineering, class imbalance handling using SMOTE, and comparison of multiple classification models.

The results show that while Random Forest achieved high accuracy, Gradient Boosting provided better overall performance in terms of F1-score and ROC-AUC, making it more suitable for imbalanced data. These findings emphasize the importance of proper preprocessing and appropriate evaluation metrics in improving model performance.

For future work, the model can be enhanced through hyperparameter tuning, cross-validation, improved interpretability, and the application of more advanced techniques or larger datasets to increase robustness and generalization.

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AUTHOR CONTRIBUTIONS

Ruksar Syed : Data Curation, Methodology, Visualization, Software, Implementation (SVM and Gradient Boosting), Formal Analysis, Writing — Original Draft, Writing — Review & Editing.

Balanjani Janjanam: Conceptualization, Background Research, Exploratory Data Analysis, Implementation (Random Forest and KNN), Validation, Writing — Review & Editing.